Design of Lead-Lag Compensation via Root-Locus

(通过根轨迹的超前-滞后补偿设计)

Ex. Typical Lead/Lag Compensator Design via Root-Locus

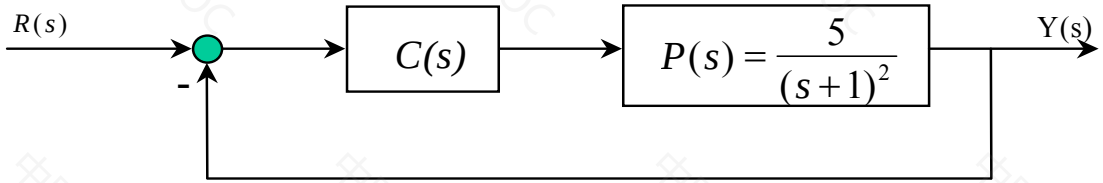
Consider the following feedback system (考虑如下反馈系统):

Design a compensator C(s) to meet the following performance specifications (设计1个控制器C(s)以满足下列性能要求):

(P1) Steady-state error to a unit step reference input must be 0.001. (单位阶跃参考输入的稳态误差为0.001)

(P2) Damping ratio of the dominant CL complex poles is around 0.5. (主导闭环复极点的阻尼比大约是0.5)

(P3) The 2% band settling time is less than 0.5 sec. (2%调整时间小于0.5秒)

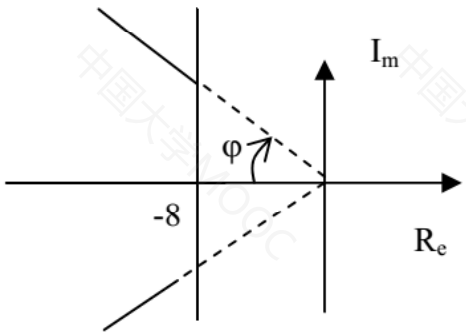


Step 0: *Obtain Performance Region on s-plane* (获得s平面中的性能域)

$$t_s=\frac{4}{\sigma}\le 0.5\rightarrow\sigma\ge 8$$

$$\tan\varphi=\frac{\omega_d}{\sigma}=\frac{\sqrt{1-\zeta^2}\,\omega_n}{\zeta\omega_n}=\frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$\zeta=0.5\rightarrow\varphi\le 60^{\circ}$$



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Step 1: *Choose a Lead or a Series of Lead Compensators to Satisfy Transient Performance Specifications, i.e., to place CL poles at desired locations or in the performance region.* **(选择1个或1系列的超前控制器来满足瞬态性能要求，也就是配置闭环极点到期望的位置上或在性能域内)**

- Single Lead Compensator **(单个超前控制器)**:

$$C_{lead} = K_c \frac{s+\beta_1}{s+\alpha_1} \qquad \beta_1 < \alpha_1$$

- Selecting Desired CL poles Locations **(选择期望闭环极点位置)**

$$p_{CL1,2} = -8 \pm 13.85j$$

satisfying ζ=0.5 and 2% t_d=0.5 sec (**满足ζ和t_d**)

$$p_{CL3} = -5 \times 8 = -40$$

5~10 times of $p_{CL1,2}$ so that the CL response will be dominated by $p_{CL1,2}$ **(5~10倍的$p_{CL1,2}$，这样闭环响应由主导极点$p_{CL1,2}$决定)**

- Desired CLCE **(期望闭环特征方程)**

$$(s-p_{CL1})(s-p_{CL2})(s-p_{CL3}) = s^3 + 56s^2 + 896s + 10233$$

- Actual CLCE **(实际闭环特征方程)**

$$1 + K_c \frac{s+\beta_1}{s+\alpha_1} \frac{5}{(s+1)^2} = 0 \Rightarrow s^3 + (2+\alpha_1)s^2 + (1+2\alpha_1+5K_c)s + \alpha_1 + 5K_c\beta_1 = 0$$

- Comparing Coefficients **(比较系数)**

$$\begin{cases} 2+\alpha_1 = 56 \\ 1+2\alpha_1+5K_c = 896 \\ \alpha_1 + 5K_c\beta_1 = 10233 \end{cases} \Rightarrow \begin{cases} \alpha_1 = 54 \\ K_c = 157.4, \\ \beta_1 = 12.93 \end{cases} \quad \therefore C_{lead}(s) = 157.4 \frac{s+12.93}{s+54}$$

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Step 2: *Check if Steady-State Performance Specifications are satisfied. If not, a lag compensator should be added, which will be done in Step 3.* **(检查稳态性能是否满足要求，如果不能，按第3步方法增加1个滞后控制器)**

- Desired Position-Error Constant **(期望位置误差常数)**:

$$\frac{1}{1+K_{pd}} = 0.001 \rightarrow K_{pd} = 999$$

- Actual Position-Error Constant for the lead compensator **(实际超前控制器的位置误差常数)**:

$$K_p = L(0) = C_{lead}(0)P(0) = 157.4 \times \frac{12.93}{54}\frac{5}{(0+1)^2} = 188.4$$

Q: Does steady-state performance specification is satisfied? (稳态性能是否满足要求)

No as K_p is smaller than the desired K_{pd} . Steady-state performance specifications is not satisfied and a lag compensator is needed to boost up the static gain **(K_p不如期望的K_{pd}小，不能满足稳态性能要求，因此需要1个滞后控制器来提升系统静态增益).**

Ex. Typical Lead/Lag Compensator Design via Root-Locus

Step 3: *Adding a lag compensator to meet the steady-state performance requirement*
(增加1个滞后控制器来满足稳态性能要求)

$$n = \frac{K_{pd}}{K_p} = \frac{999}{188.4} = 5.3 \qquad C_{lag}(s) = \frac{s + n\alpha_I}{s + \alpha_I}$$

Choose

$$n\alpha_I = \frac{1}{20}\left|\mathrm{Re}[p_{CL1,2}]\right| = \frac{1}{20} \times 8 = 0.4 \Rightarrow \alpha_I = \frac{0.4}{5.3} = 0.0755$$

$$C(s) = C_{lead}(s)\, C_{lag}(s) = 157.4 \frac{s + 12.93}{s + 54} \frac{s + 0.4}{s + 0.0755}$$

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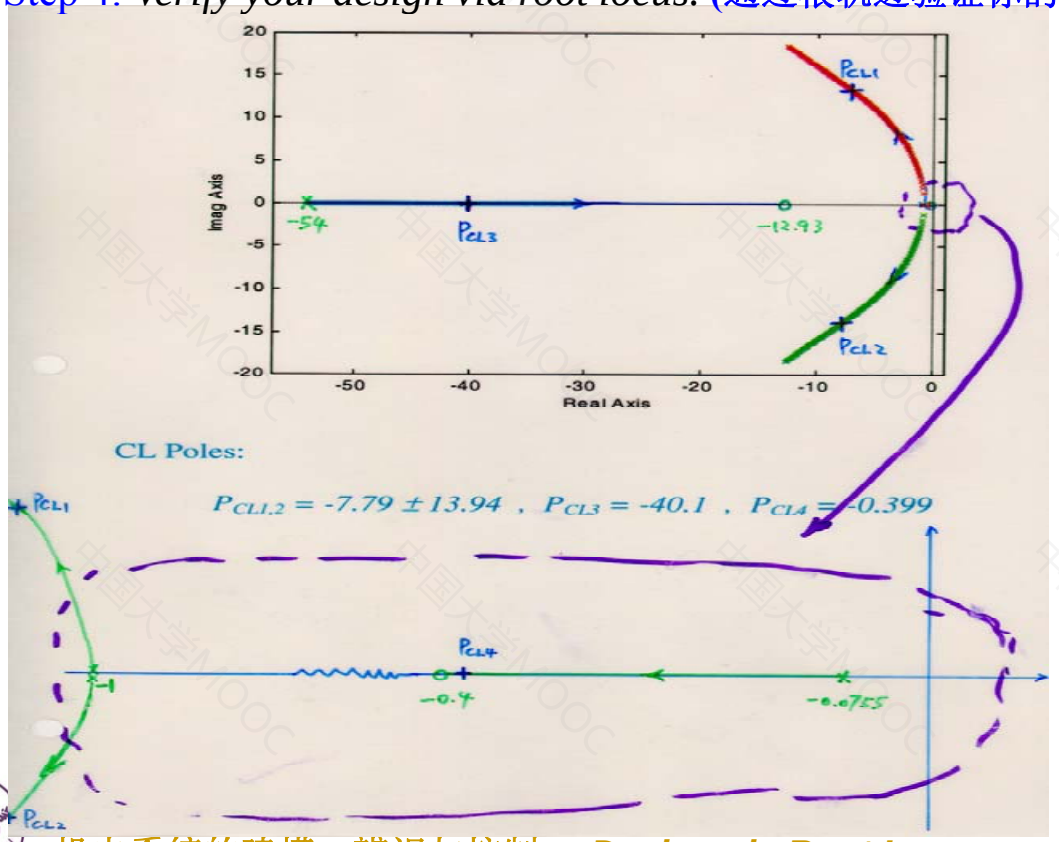
Step 4: *Verify your design via root locus.* (通过根轨迹验证你的设计)



$$L(s) = P(s)C(s) = \frac{5}{(s+1)^2} 157.4 \frac{s+12.93}{s+54} \frac{s+0.4}{s+0.0755}$$

The last CL pole p_{CL4} is very close to the zero of the lag compensator, which remains as one of the CL zeros! (最终闭环极点非常接近滞后控制器的零点，控制器零点仍然是闭环零点之一)

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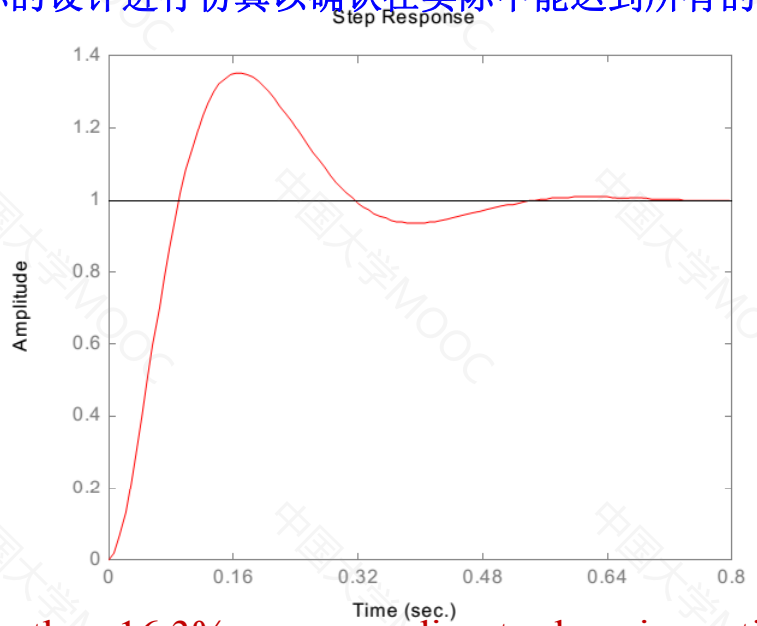
Step 5: *Simulate your design to make sure all performance specifications are actually met. If not, iterate on the design* (对你的设计进行仿真以确认在实际中能达到所有的性能要求。如果不能，重新设计)



The actual OS% is larger than 16.3% corresponding to damping ratio of 0.5. This discrepancy is due to the CL zero contributed by the zeros of the lead-lag compensator. The 2% settling time is satisfied. **(实际超调量比0.5阻尼比对应的16.3%要大些，这种差异是由于超前-滞后补偿器的零点带来的闭环零点造成的，而2%调整时间满足要求了。)**

Summary

- **Structure of Lead-Lag Compensation** (超前-滞后补偿器结构)
- **Design of Lead Compensation** (超前补偿器设计)
- **Design of Lag Compensation** (滞后补偿器设计)
- **Typical Lead/Lag Compensator Design via Root-Locus** (通过根轨迹的典型超前-滞后补偿器设计)